

Tenacious-Bench v0.1 & Preference-Tuned Judge

To: CEO & CFO, Tenacious Consulting · From: Engineering · 2 May 2026 · Page 1 of 2

Executive Summary

We built **Tenacious-Bench v0.1**, a 266-task benchmark targeting four failure modes τ^2 -Bench retail cannot grade — wrong-segment pitching (D06), signal overclaiming (I01–I03), injection attacks (E01–E05), and trajectory drift — and trained a DPO preference-tuned judge (LoRA, Qwen 2.5-0.5B) that achieves **74.0% accuracy on the sealed held-out partition (95% CI [62%, 86%])**, a **+26 pp lift over the deterministic rule evaluator ($p = 0.0127$, paired bootstrap $n = 10,000$)**. The judge functions as a rejection-sampling gate: emails flagged FAIL are routed to human review, blocking brand-damaging wrong-segment pitches at a cost of **\$0.0002 per task** and ~2 s latency, versus \$0.00 and <1 ms for the rule evaluator alone. We recommend **deploy with caveat**: wire the judge into production as an async rejection-sampling layer, cap daily gated volume at 50 emails, and retrain on a balanced 50/50 chosen-PASS / chosen-FAIL preference set using Qwen 2.5-3B before full rollout.

Headline Lift — Delta A

Judge	Accuracy	95% CI	p vs rule
DPO trained judge	74.0%	[62%, 86%]	0.0127 ✓
Rule evaluator (baseline)	48.0%	[34%, 62%]	—
Prompt judge — zero-shot qwen3-8b	22.0%	[12%, 34%]	—

Delta B — Prompt-Engineered Baseline (Honest Report)

The zero-shot prompt judge predicted **PASS for all 50 held-out tasks** (100% PASS bias), achieving only 22% accuracy — 0 failures correctly caught. This confirms that prompt engineering alone cannot replicate preference training: the base model must see explicit contrastive examples of grounded compliance versus keyword-passing hallucination to overcome the verbosity/politeness bias (Zheng et al., 2023). The trained judge's +52 pp lift over the prompt baseline (74% vs 22%, $p < 0.0001$) directly validates the Path B choice.

Cost per Task & Production Implication

Component	\$/task	Latency	Accuracy
No judge (current)	\$0.000	0 ms	— (blind)
Rule evaluator only	\$0.000	<1 ms	48%
Prompt judge zero-shot	\$0.000135	~1.5 s	22%
DPO trained judge ✓	\$0.0002	~2 s (T4)	74%

A +26 pp accuracy gain at **\$0.0002/task** is cost-Pareto dominant: the marginal judge cost is <1% of the estimated cost of one misdirected email causing prospect churn. The 0.5B backbone's FAIL bias (18% PASS recall, 2/11 GT-PASS correctly passed) means the judge over-blocks until retrained on a balanced preference set with Qwen 2.5-3B or larger.

Coverage Gaps, Ground Truth Critique & Kill-Switch

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Four Failure Modes Tenacious-Bench v0.1 Does Not Capture

1. Multi-turn conversation coherence.

All 266 tasks evaluate a single email in isolation. A judge excelling here can still fail a 3-email thread where tone shifts from consultative to aggressive as the agent loses its grounded signal anchor across turns. **v0.2 fix:** 30–50 multi-turn trajectory tasks with per-turn rubric scoring and a tone-drift dimension penalising segment-inconsistent escalation.

2. Live prospect emotional reception.

The rubric scores text compliance, not how a real CTO perceives the message. An email passing all checks can feel condescending to a VP who deliberately chose not to adopt the capability being pitched. **v0.2 fix:** ICP-persona tone-panel probes using LLM-as-prospect scoring on perceived-value and perceived-relevance scales, calibrated against the 12 hand-labeled 'good' drafts in Style Guide v2.

3. Bench-capacity temporal drift.

Tasks use static bench summaries. Bench availability changes daily; a 24-hour delay between bench check and send creates an over-commitment window the benchmark never simulates. A judge trained on static snapshots will not flag commitments that became false by send time. **v0.2 fix:** time-variant bench-state injection paired with a temporal-staleness failure category.

4. Non-English and multilingual outreach.

All 266 tasks are English-only. Tenacious operates in East Africa where Amharic, Swahili, and French outreach may be required; tone compliance rules differ across languages and cultures. **v0.2 fix:** ≥30 tasks per language with culturally-calibrated tone rubrics and separate per-language IRA protocols.

Public-Signal Lossiness in Ground Truth

25 of the 50 sealed held-out tasks are LLM-synthesis mode and carry a **systematic FAIL labeling artifact**: the synthetic generator defaults to a FAIL verdict with empty failure categories (`fail_cats=[]`) when task intent is ambiguous instead of escalating to human review. The rule evaluator's accuracy on LLM-synthesis tasks is 36% versus 58–62% on programmatic and hand-authored tasks — a 22–26 pp gap inflating both apparent held-out difficulty and the measured Delta A. Concretely: the +26 pp lift may partly reflect the trained judge learning the generator's labeling convention rather than true semantic alignment failure detection. **v0.2 fix:** mandatory human spot-check on 20% of LLM-synthesis tasks before held-out sealing; reject any task with `fail_cats=[]`.

Honest Unresolved Training Failure

The DPO judge over-corrected from the prompt judge's 100% PASS bias to a near-100% **FAIL bias**: 48 of 50 held-out tasks predicted FAIL against a GT FAIL rate of 78% (39/50). On the 11 GT-PASS tasks only 2 were correctly passed (18% PASS recall). Root cause: the 279 preference pairs contained three rejection tiers (blatant, subtle, hard-negative FAIL) but no explicit chosen-PASS tier with positive compliant examples — the model learned 'what FAIL looks like' without a matching signal for PASS. The fix is not more training data of the same type; it is reconstructing the preference set with a 50/50 chosen-PASS / chosen-FAIL balance before the next training run.

Kill-Switch Trigger Conditions

- **Monthly calibration accuracy drops below 60%** on a 30-task dev-partition spot-check. Below this threshold the judge provides less signal than a coin flip.
- **False-negative rate on GT-PASS tasks exceeds 90%** (current: 82%). At 90% the judge suppresses >9 in 10 legitimate emails — over-blocking harm exceeds brand-protection benefit.
- **Per-task inference cost exceeds \$0.01** (e.g. GPU failure forcing CPU fallback). At 50× current cost the judge is no longer cost-Pareto dominant; revert to rule evaluator (\$0.00/task).
- **Adapter hash diverges from pinned revision** in `model_card.md`. Halt deployment and re-verify adapter against `training/training_run.log` before re-enabling.

Revert trigger: any single condition above → switch immediately to rule evaluator; open a GitHub issue with the calibration log attached; do not re-enable until a corrected adapter passes a fresh 50-task calibration run.